Technical Reflection

- Making the app console-based was probably not a smart decision, as the app cannot receive a message while one is typing, it creates a MESS!
- The server IP must be the same on all client files, and it must be the one of the servers currently running.
- Running the Client.py file with the wrong IP address creates a “Time-Out Error”
- TCP ensured safe delivery, but it did not ensure that text message is received in full:
- Because of that, we had to implement message framing
- The Redis queue mimic behavior was not working as users would receive messages when they were offline, but when they go back online, they don’t receive anything.
- The code implementation was updated to account for that.
- Images are still not being received if they were sent while the user was offline, but a message is sent to let them know that an image was received.
- The server is not encrypted as it can read user messages. To be updated in stage 3.
- Stage 3 will use GUI to implement the app.
- The app will make use of PostgreSQL, Redis server and Google Cloud Storage to accurately implement the intended behavior.
- To implement the hybrid system so that files can use UDP to be transferred between offline users and TCP for offline users.
- Check for password instead of allowing for every password in stage 3.

Some evidence:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka
.ms/PSWindows

PS C:\Users\trace\UCT\CSC3002F\C00NECTED> py ARCPServer.py
Server listening on 196.47.192.177:50000...
[REGISTERED] tracey at 196.47.192.177:57160
[RECEIVED] From tracey: SEND:kb:Hey
[REGISTERED] omphile at 196.47.192.177:49167
[RECEIVED] From omphile: SEND:tracey:hi
[RECEIVED] From tracey: SEND:omphile:hi how are you?
[RECEIVED] From omphile: SEND:tracey:I am good and how are you?
```

```
Windows PowerShell
PS C:\Users\trace\UCT\CSC3002F\C00NECTED> py Client.py
Enter username: tracey
User not found. Do you want to register? (yes/no): yes
Enter a unique username: tracey
Enter new password: 1234

[SYSTEM] Registration successful! Welcome to the Chat.
[UDP] Listening for P2P media on unique port 57160...
>> SEND:kb:Hey
[DEBUG] Sending message to kb: Hey

QUEUED_OFFLINE>>

[omphile]: hi
SEND:omphile:hi how are you?
[DEBUG] Sending message to omphile: hi how are you?
>>
[omphile]: I am good and how are you?
```

```
Windows PowerShell

Install the latest PowerShell for new features and improvements! https://aka
.ms/PSWindows

PS C:\Users\trace\UCT\CSC3002F\C00NECTED> py Client.py
Enter username: omphile
User not found. Do you want to register? (yes/no): yes
Enter a unique username: omphile
Enter new password: 1234

[SYSTEM] Registration successful! Welcome to the Chat.
[UDP] Listening for P2P media on unique port 49167...
>> SEND:tracey:hi
[DEBUG] Sending message to tracey: hi
>>
[tracey]: hi how are you?
SEND:tracey:I am good and how are you?
[DEBUG] Sending message to tracey: I am good and how are you?
>>
```

```
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Install the latest PowerShell for new features and improvements! https://aka
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PS C:\Users\trace\UCT\CSC3002F\C00NECTED> py ARCPServer.py
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[RECEIVED] From omphile: SEND:tracey:hi
[RECEIVED] From tracey: SEND:omphile:hi how are you?
[RECEIVED] From omphile: SEND:tracey:I am good and how are you?
[ERROR] with ('196.47.192.177', 53690): [WinError 10054] An existing connect
ion was forcibly closed by the remote host
[RECEIVED] From tracey: SEND:omphile:I am good.
[ERROR] with ('196.47.192.177', 53940): [WinError 10038] An operation was at
tempted on something that is not a socket
[REGISTERED] omphile at 196.47.192.177:58243
[RECEIVED] From omphile: SEND:tracey:Hey
[ERROR] with ('196.47.192.177', 50370): [WinError 10038] An operation was at
tempted on something that is not a socket
```
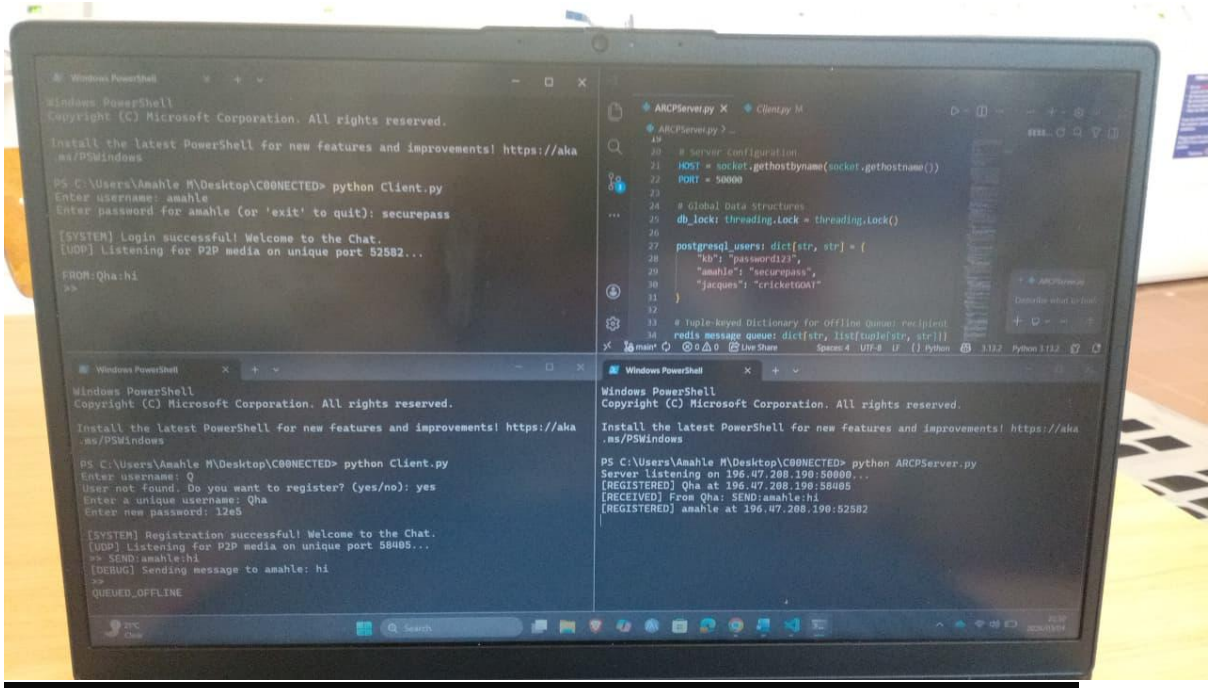

```
Windows PowerShell
[DEBUG] Sending message to omphile: Get Amahle on App
>> SEND:omphile:Proteas mara
[DEBUG] Sending message to omphile: Proteas mara
Traceback (most recent call last):
  File "C:\Users\trace\UCT\CSC3002F\C00NECTED\Client.py", line 314, in <modu
le>
    start_client()
    ~~~~~~~~~~~~^^
  File "C:\Users\trace\UCT\CSC3002F\C00NECTED\Client.py", line 298, in start
_client
    send_framed_msg(tcp_sock, f"SEND:{recipient}:{data}", 'D')
    ~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\trace\UCT\CSC3002F\C00NECTED\Client.py", line 132, in send_
framed_msg
    sock.sendall(header + data)
    ~~~~~~~~~~~~^^^^^^^^^^^^^^^
ConnectionAbortedError: [WinError 10053] An established connection was abort
ed by the software in your host machine
PS C:\Users\trace\UCT\CSC3002F\C00NECTED>
```

```
Windows PowerShell

[SYSTEM] Registration successful! Welcome to the Chat.
[UDP] Listening for P2P media on unique port 49167...
>> SEND:tracey:hi
[DEBUG] Sending message to tracey: hi
>>
[tracey]: hi how are you?
SEND:tracey:I am good and how are you?
[DEBUG] Sending message to tracey: I am good and how are you?
>> EXIT
PS C:\Users\trace\UCT\CSC3002F\C00NECTED> py Client.py
Enter username: omphile
Enter password for omphile (or 'exit' to quit): 1234

[SYSTEM] Login successful! Welcome to the Chat.
[UDP] Listening for P2P media on unique port 58243...
>> SEND:tracey:Hey
[DEBUG] Sending message to tracey: Hey
>>
```




```
>> EXIT
PS C:\Users\KaraboG\Desktop\Networking Stage 2> python Clier
Enter username: kb
Enter password for kb (or 'exit' to quit): password123

[SYSTEM] Login successful! Welcome to the Chat.
[UDP] Listening for P2P media on unique port 51480...

--- Commands ---
SEND:<user>:<message>              - Direct Message
CREATE_GROUP:<group_name>          - Create a new group
ADD_TO_GROUP:<group_name>:<user>   - Add a user to a group
LEAVE_GROUP:<group_name>           - Leave a group
SEND_GROUP:<group_name>:<msg>      - Message a group
SENDFILE:<user/group>:<filepath>   - P2P Media Transfer
COMMANDS                           - Show this help menu
EXIT                               - Disconnect


[2026-03-05 02:02:32] [Karabo (DM)]: KB
>>
```